

Bharth K S

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Summary

Applied AI Engineer with an MSc in Robotics and expertise in LLM integration, adversarial safety testing, and autonomous perception systems. Skilled in Python, C++, and state-of-the-art frameworks including Anthropic Claude, OpenAI API, and ROS 2. Proven track record in building GenAI interaction workflows and robotic automation. Seeking to contribute practical software engineering skills to scalable AI deployments as an Applied AI Engineer.

Experiences

Startup Ecosystem Associate Dec 2025 — Present
Venture Cafe UK

- Identified for **AI model implementation**, ensuring technical specifications aligned with business objectives to accelerate project delivery.
- Presented technical **AI product demonstrations** for diverse client groups, articulating the value of **AI/ML workflows**.
- Managed AI-enabled service operations by resolving complex technical queries and maintaining system reliability, which increased service accuracy scores and client satisfaction.

Production Engineering Intern Jun 2025 — Jul 2025
Hyundai Motors Pvt. Ltd. India

- Automated quality validation processes for **robotic production systems** across multiple assembly lines, saving **10+ technician-hours per day** and improving system consistency.
- Analysed **operational performance datasets** to identify efficiency trends, preventing an average of **4 production line stoppages** per week through continuous process optimisation efforts.
- Generated structured production reports to track throughput and quality metrics, reducing **5+ hours of manual data consolidation** weekly while ensuring strict compliance.

Projects

Adversarial Safety In Embodied AI – LLM Attack & Defence System

- Designed an adversarial attack and defense framework for LLM-powered agents in AI2-THOR virtual ecosystems by integrating **Anthropic Claude** and **Ollama** models to perform prompt injection testing.

Gen AI Integration with Robotic Systems

- Integrated **advanced language models** with robotic platforms to enable real-time **natural language understanding** and applied prompt engineering to optimise behavioural control and decision-making processes.

Rescue Agent (Hackathon Project)

- Developed a **Streamlit** application using **AWS Strand** and **Claude 3.5 Opus** to manage food redistribution workflows, validated by an internal pilot evaluation using 3 driver and 50 food-specification datasets to ensure system reliability and accuracy.

Education

MSc Robotics Sep 2025 — Present
Heriot-Watt University Edinburgh, United Kingdom

Robotics, Autonomous Systems, Machine Learning

Bachelor of Engineering (B.E.) in Mechatronics Engineering Jun 2021 — 2025
Kongu Engineering College India

Control Systems, Embedded Systems, Mechatronics Design

Skills

AI & Computer Vision : Python, C++, C, MATLAB, OpenAI API, Anthropic Claude, Ollama, Hugging Face, Prompt Engineering, LLM Fine-Tuning, RAG, Machine Learning, AI Safety, Responsible AI, Adversarial Testing, Hallucination Mitigation, OpenCV

Robotics & IoT : ROS 1, ROS 2, SLAM, Sensor Fusion, Kalman Filtering, Autonomous Navigation, Image Processing, Visual Perception Systems, ESP32, MQTT, Node-RED, ThingSpeak

Developer Tools & Software : Docker, Git, Jupyter Notebook, VS Code, AI2-THOR, Streamlit, SolidWorks, Autodesk CAD, Ansys, LabVIEW, RoboStudio